

KUNAL HEMANT PASAD

Machine Learning Engineer | Deep Learning | Computer Vision | NLP | Generative AI

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EDUCATION

Rutgers, the State University of New Jersey, New Brunswick, United States

Expected: May 2027

Master of Science in Computer Science

GPA: 3.83/4.0

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Distributed Systems

Sardar Patel Institute of Technology, Mumbai, India

Dec 2021 – May 2025

Bachelor of Technology in Computer Engineering

GPA: 8.62/10

Relevant Coursework: Data Structures & Algorithms, Probability, Statistics, Database Systems

TECHNICAL SKILLS

Languages: Python, SQL

Deep Learning: PyTorch, CNNs, Neural Networks, Model Training, Optimization, Transfer Learning

ML & AI: scikit-learn, LLMs, HuggingFace Transformers, QLoRA, RAG

Computer Vision: Image Processing, Spectrogram Analysis, Feature Extraction (MFCC), CNN Architectures

Data: Pandas, NumPy, Feature Engineering, Time-Series Modeling

Deployment: FastAPI, AWS (EC2, S3, Lambda), Docker

Concepts: Model Evaluation, Explainable AI, Multimodal Learning, Prompt Engineering

PROFESSIONAL EXPERIENCE

EternIQ

Jan 2025 – May 2025

Machine Learning Engineer Intern

Mumbai, India

- Engineered scalable LLM inference pipelines enabling structured reasoning and code generation across production systems
- Designed prompt engineering strategies reducing hallucination rates by 25% improving model output reliability
- Developed evaluation framework analyzing reasoning accuracy across 1K+ outputs improving consistency and enhancing model performance
- Built data pipelines processing 100K+ records supporting model training, validation, and large-scale experimentation workflows

Mukesoftware IT Consultants

Jul 2023 – Feb 2024

Full Stack Web Developer Intern

Mumbai, India

- Developed backend APIs supporting data-driven systems and efficient processing pipelines across applications
- Optimized SQL queries and indexing reducing response latency by 30% improving overall system performance significantly
- Processed over 50K relational records improving storage efficiency and enabling faster analytical query execution

PROJECTS

InstantStory – Multimodal LLM Story Generation System | PyTorch, HuggingFace, QLoRA, FastAPI

Mar 2026

- Engineered LLM pipeline curating 13K+ samples improving narrative coherence by 22% using structured model training workflows
- Fine-tuned Mistral-7B using QLoRA reducing memory usage by 60% while maintaining high-quality generation outputs
- Optimized inference latency by 25% through batching strategies and efficient prompt construction techniques

Multilingual Sports QA Benchmark (EMNLP 2025) | LLMs, Multilingual NLP

Nov 2025

- Developed benchmark dataset with 33K+ samples across 11 languages enabling multilingual reasoning evaluation tasks
- Designed multi-hop reasoning pipelines generating 30K+ samples for compositional reasoning evaluation
- Analyzed cross-lingual performance identifying 18% degradation across models under multilingual conditions

MindSync – AI Knowledge Assistant | RAG, FastAPI, AWS

Oct 2025

- Architected RAG system reducing hallucinations by 25% using retrieval-based grounding over unstructured document corpora
- Built embedding and vector search pipeline improving response latency by 30% across real-time inference queries
- Deployed scalable backend on AWS enabling distributed processing and asynchronous data ingestion workflows

Deepfake Guard – Audio Deepfake Detection System | PyTorch, CNNs, MFCC, Explainable AI

Jun 2025

- Developed CNN-based deep learning model achieving 93% accuracy on 15K+ audio samples using spectrogram features
- Engineered MFCC-based feature extraction pipeline transforming raw audio into structured inputs for training
- Applied SHAP and Grad-CAM improving interpretability and robustness by 18% through targeted augmentations